

Chat Link: <https://parley.mit.edu/share/Tx-5hVIsaMi8p1INgPCJq>

```
global_data={
  "attribute_1": "total_cost",
  "attribute_2": "material_continuity",
  "attribute_3": "electrical_conductivity_pct_iacs",
  "attribute_4": "insulation_temperature_rating_wet_c",
  "attribute_5": "installation_joining_compatibility_score",

  "direction_attribute_1": "negative",
  "direction_attribute_2": "positive",
  "direction_attribute_3": "positive",
  "direction_attribute_4": "positive",
  "direction_attribute_5": "positive",

  "attribute_1_min": 0.50,
  "attribute_1_max": 150.00,
  "attribute_2_min": 0,
  "attribute_2_max": 1,
  "attribute_3_min": 30,
  "attribute_3_max": 105,
  "attribute_4_min": 50,
  "attribute_4_max": 130,
  "attribute_5_min": 1,
  "attribute_5_max": 10,

  "copper_price_per_ton": 9500,
  "aluminum_price_per_ton": 2500,
  "copper_price_per_kg": 9.5,
  "aluminum_price_per_kg": 2.5
}
```

```
regional_data=[
  {
    "region": "United States",
    "copper_product_market_share": 0.75,
    "aluminum_product_market_share": 0.25,
    "weight_attribute_1": 3,
    "weight_attribute_2": 4,
    "weight_attribute_3": 5,
    "weight_attribute_4": 4,
    "weight_attribute_5": 3
  },
]
```

```

{
  "region": "China",
  "copper_product_market_share": 0.55,
  "aluminum_product_market_share": 0.45,
  "weight_attribute_1": 5,
  "weight_attribute_2": 3,
  "weight_attribute_3": 3,
  "weight_attribute_4": 3,
  "weight_attribute_5": 2
},
{
  "region": "India",
  "copper_product_market_share": 0.40,
  "aluminum_product_market_share": 0.60,
  "weight_attribute_1": 5,
  "weight_attribute_2": 3,
  "weight_attribute_3": 3,
  "weight_attribute_4": 2,
  "weight_attribute_5": 2
},
{
  "region": "Europe",
  "copper_product_market_share": 0.70,
  "aluminum_product_market_share": 0.30,
  "weight_attribute_1": 3,
  "weight_attribute_2": 4,
  "weight_attribute_3": 4,
  "weight_attribute_4": 5,
  "weight_attribute_5": 4
}
]

```

```

product_data=[
  {
    "region": "United States",
    "dominant material": "cu",
    "product": "Commercial Trunk Line – 250 MCM Cu THHN/THWN-2, per meter",
    "copper_kg": 1.14,
    "aluminum_kg": 0.00,
    "total_mass_kg": 1.30,
    "non_material_cost_per_unit": 5.00,
    "attribute_2_value": 1.0,
    "attribute_3_value": 100,

```

```
"attribute_4_value": 75,
"attribute_5_value": 9
},
{
  "region": "United States",
  "dominant material": "al",
  "product": "Commercial Trunk Line – 350 MCM Al XHHW-2, per meter",
  "copper_kg": 0.00,
  "aluminum_kg": 0.48,
  "total_mass_kg": 0.70,
  "non_material_cost_per_unit": 5.50,
  "attribute_2_value": 0.0,
  "attribute_3_value": 61,
  "attribute_4_value": 90,
  "attribute_5_value": 4
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "Commercial Trunk Line – 250 MCM Cu THWN-2 / GB equiv., per meter",
  "copper_kg": 1.14,
  "aluminum_kg": 0.00,
  "total_mass_kg": 1.30,
  "non_material_cost_per_unit": 4.00,
  "attribute_2_value": 1.0,
  "attribute_3_value": 100,
  "attribute_4_value": 75,
  "attribute_5_value": 9
},
{
  "region": "China",
  "dominant material": "al",
  "product": "Commercial Trunk Line – 350 MCM Al XHHW-2 / GB equiv., per meter",
  "copper_kg": 0.00,
  "aluminum_kg": 0.48,
  "total_mass_kg": 0.70,
  "non_material_cost_per_unit": 4.50,
  "attribute_2_value": 0.0,
  "attribute_3_value": 61,
  "attribute_4_value": 90,
  "attribute_5_value": 4
},
{
  "region": "India",
```

```

    "dominant material": "cu",
    "product": "Commercial Trunk Line – 250 MCM Cu THWN-2 / IS equiv., per meter",
    "copper_kg": 1.14,
    "aluminum_kg": 0.00,
    "total_mass_kg": 1.30,
    "non_material_cost_per_unit": 3.50,
    "attribute_2_value": 1.0,
    "attribute_3_value": 100,
    "attribute_4_value": 75,
    "attribute_5_value": 9
  },
  {
    "region": "India",
    "dominant material": "al",
    "product": "Commercial Trunk Line – 350 MCM Al XHHW-2 / IS equiv., per meter",
    "copper_kg": 0.00,
    "aluminum_kg": 0.48,
    "total_mass_kg": 0.70,
    "non_material_cost_per_unit": 4.00,
    "attribute_2_value": 0.0,
    "attribute_3_value": 61,
    "attribute_4_value": 90,
    "attribute_5_value": 4
  },
  {
    "region": "Europe",
    "dominant material": "cu",
    "product": "Commercial Trunk Line – 250 MCM Cu XLPE / IEC 60502-1, per meter",
    "copper_kg": 1.14,
    "aluminum_kg": 0.00,
    "total_mass_kg": 1.30,
    "non_material_cost_per_unit": 6.50,
    "attribute_2_value": 1.0,
    "attribute_3_value": 100,
    "attribute_4_value": 90,
    "attribute_5_value": 9
  },
  {
    "region": "Europe",
    "dominant material": "al",
    "product": "Commercial Trunk Line – 350 MCM Al XLPE / IEC 60502-1, per meter",
    "copper_kg": 0.00,
    "aluminum_kg": 0.48,
    "total_mass_kg": 0.70,

```

```
"non_material_cost_per_unit": 6.00,  
"attribute_2_value": 0.0,  
"attribute_3_value": 61,  
"attribute_4_value": 90,  
"attribute_5_value": 4  
}  
]
```